

day 08 | 250915

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: data = np.loadtxt('../data/150407-02.dat', skiprows=1)
```

```
In [7]: data[1]
```

```
Out[7]: array([28., 17., 23., ..., 8., 12., 0.])
```

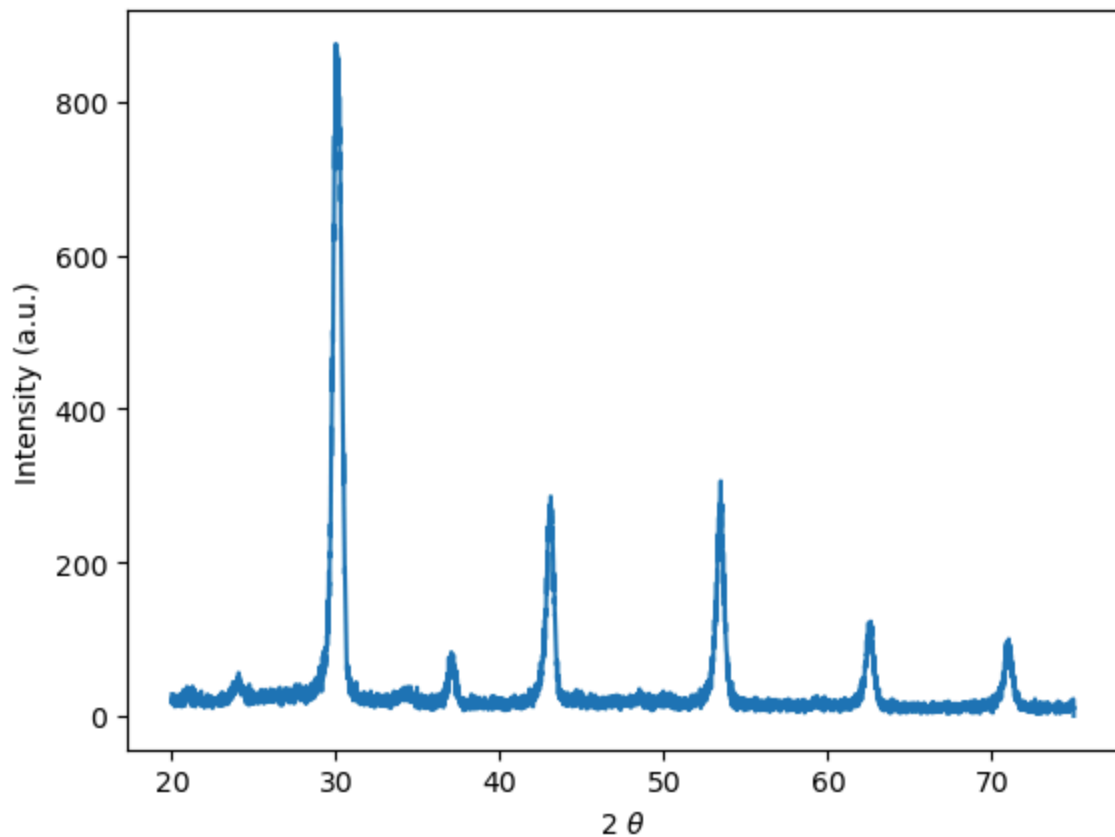
```
In [4]: data = data.T
```

```
In [12]: fig0 = plt.figure()
ax0 = fig0.add_subplot()

ax0.plot(data[0], data[1])

ax0.set_xlabel(r'2 $\theta$')
ax0.set_ylabel('Intensity (a.u.)')
```

```
Out[12]: Text(0, 0.5, 'Intensity (a.u.)')
```



```
In [13]: fig0.savefig('../figs/150407-02-plot.png')
```

```
In [14]: data1 = pd.read_csv('../data/170330-00.mpt', header=55, sep='\t', encoding='')
```

```
In [15]: data1
```

```
Out[15]:
```

	freq/Hz	Re(Z)/Ohm	-Im(Z)/Ohm	Z /Ohm	Phase(Z)/deg	time/s	<Ewe>/V
0	7.000024e+06	895.75378	147.372860	907.79602	-9.342825	2.520596	0.0003
1	5.551806e+06	978.20587	77.754303	981.29120	-4.544694	3.437591	-0.0000
2	4.403234e+06	966.23254	76.227158	969.23468	-4.510785	3.546738	-0.0000
3	3.492272e+06	968.32733	64.608551	970.48035	-3.817220	3.656734	-0.0000
4	2.769775e+06	974.00806	62.624184	976.01923	-3.678788	3.866602	-0.0000
...	...	...	...	...	...	...	...
64	2.526678e+00	5487.50390	466.635010	5507.30860	-4.860507	17.487966	0.0003
65	2.003207e+00	5506.98240	372.655120	5519.57670	-3.871279	18.595106	0.0003
66	1.589842e+00	5523.89450	295.391200	5531.78710	-3.060986	19.301461	0.0003
67	1.261711e+00	5530.05370	236.371870	5535.10300	-2.447512	20.187411	0.0003
68	1.000061e+00	5536.58640	188.091290	5539.78030	-1.945728	21.301483	0.0003

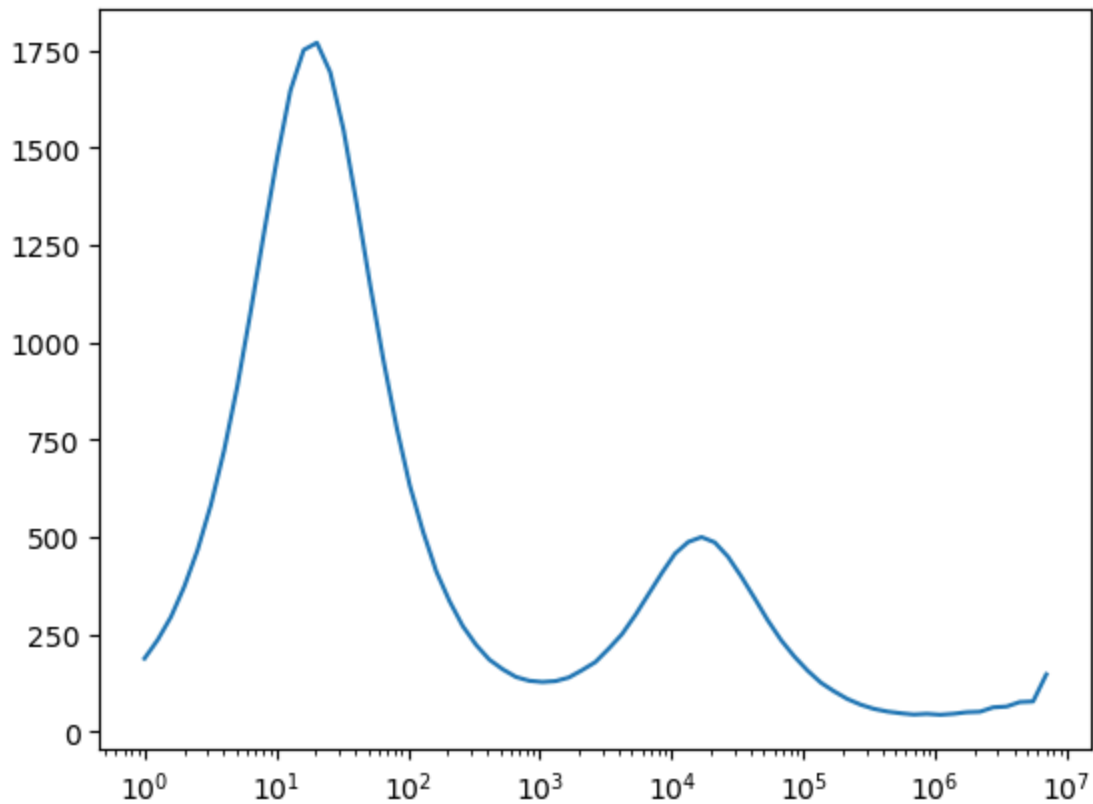
69 rows × 8 columns

```
In [16]: data1.columns
```

```
Out[16]: Index(['freq/Hz', 'Re(Z)/Ohm', '-Im(Z)/Ohm', '|Z|/Ohm', 'Phase(Z)/deg',  
               'time/s', '<Ewe>/V', '<I>/mA', 'Cs/μF', 'Cp/μF', 'cycle number',  
               'I Range', '|Ewe|/V', '|I|/A', 'Re(Y)/Ohm-1', 'Im(Y)/Ohm-1',  
               '|Y|/Ohm-1', 'Phase(Y)/deg'],  
              dtype='object')
```

```
In [19]: fig1 = plt.figure()  
ax1 = fig1.add_subplot()  
  
ax1.semilogx(data1['freq/Hz'], data1['-Im(Z)/Ohm'])
```

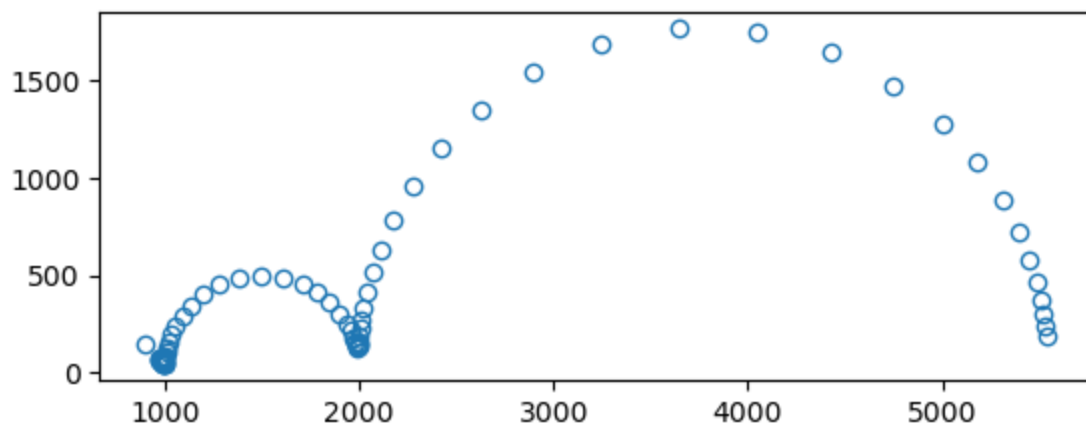
```
Out[19]: [ <matplotlib.lines.Line2D at 0x722b6b1e1b20>]
```



```
In [22]: fig2 = plt.figure()
ax2 = fig2.add_subplot(aspect='equal')

ax2.plot(data1['Re(Z)/Ohm'], data1['-Im(Z)/Ohm'], 'o', mfc='none')
```

```
Out[22]: [<matplotlib.lines.Line2D at 0x722b635918b0>]
```



homework problem exercise 3.1

This one is fine until you get to the rolling average part. Thankfully there is a library for that. Use pandas to load the data, then use the `rolling().mean()` method to implement. See below.

```
In [49]: data3 = pd.read_csv('../data/sunspots.txt', sep='\t', names=['time', 'spots'])
```

```
In [50]: data3
```

```
Out[50]:
```

	time	spots
0	0	58.0
1	1	62.6
2	2	70.0
3	3	55.7
4	4	85.0
...	...	...
3138	3138	16.1
3139	3139	19.6
3140	3140	25.2
3141	3141	23.5
3142	3142	21.6

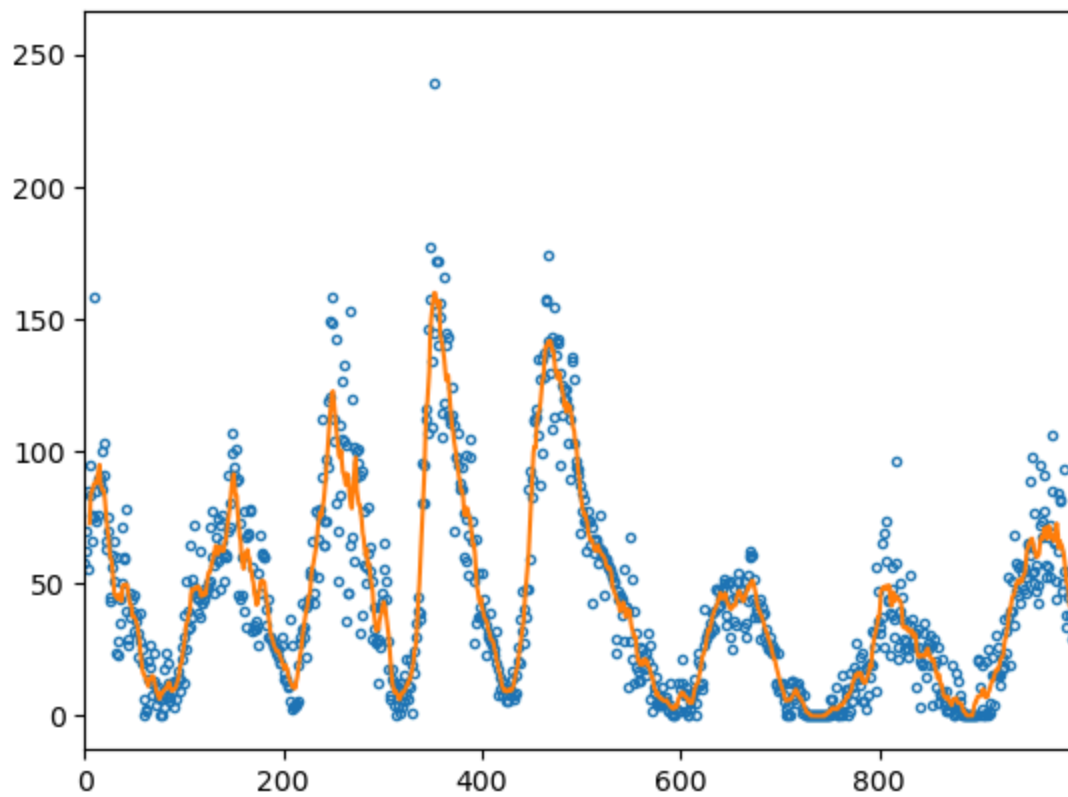
3143 rows × 2 columns

```
In [55]: fig3 = plt.figure()
ax3 = fig3.add_subplot()

ax3.plot(data3['time'], data3['spots'], '.', mfc='none')
ax3.plot(data3['time'], data3['spots'].rolling(10, center=True).mean())

ax3.set_xlim(0, 1000-1)
```

```
Out[55]: (0.0, 999.0)
```



In []:

In []: